

EncoderX Remote Internship

Track: Artificial Intelligence & Machine Learning — Batch: 02

Task 2: Sentiment Analysis Using NLP

Week 02 Final Project Submission

Required Submission Links

Official Project Deliverables

- **GitHub Repository Link:**
<https://github.com/Hazem-Emam-404/EncoderX-Internship/tree/main/Assignment-2>
(Contains complete source code, streamlined `notebook.ipynb`, `README.md`, and `requirements.txt`)
- **LinkedIn Post Link:**
<https://www.linkedin.com/in/hazem-emam-645436272/>
(Includes project summary, findings, tagging @EncoderX, and official hashtags)
- **Video Demonstration Link (3–5 min):**
<https://drive.google.com/file/d/1hShQcmSKZGZX2rBYVed-4ZQIat0QLZMJ/view?usp=sharing>
(Screen walkthrough covering data preprocessing, TF-IDF vectorization, model benchmark, and live sample inference)

Executive Summary & Methodology

- **Dataset:** Amazon Fine Food Reviews (525k+ raw reviews). Binary classification defined as Positive (**Score** > 3) vs. Negative (**Score** < 3). Neutral 3-star reviews were removed to eliminate classification ambiguity.
- **Class Balancing:** The severe 5:1 positive class skew was balanced by downsampling positive reviews to 100,000, combined with all 82,037 negative reviews (total: 182,037 reviews, 80/20 stratified split).
- **Text Preprocessing:** Implemented HTML tag stripping, regex noise removal, lowercasing, tokenization, **negation-preserving stopwords filtering** (ensuring words like "not" and "never" remain to retain polarity), and WordNet lemmatization.
- **Feature Extraction:** Transformed cleaned text into numerical vectors using **TF-IDF** with unigrams and bigrams (`ngram_range=(1,2)`), capped at the top 10,000 features.

Model Evaluation Benchmark

All models were evaluated on the held-out test set of **36,408 reviews**:

Model	Accuracy	Precision	Recall	F1-Score	Deployment Evaluation
Random Forest	90.88%	90.97%	90.61%	90.76%	Highest raw score; computationally intensive.
Logistic Regression	90.70%	90.62%	90.59%	90.60%	Production Champion: Sub-ms latency, lightweight.
Multinomial Naive Bayes	87.71%	87.89%	87.29%	87.51%	Fast baseline; lower negative recall.

Business Applications & Recommendations

- **Production Champion: Logistic Regression** is selected for production deployment due to its high F1-score (90.60%), minimal memory footprint, and sub-millisecond inference time.
- **Business Value:** Enables automated customer support ticket prioritization (triage), proactive product defect alerts for QA teams, and real-time social media brand reputation tracking.
- **Future Improvements:** Fine-tuning transformer models (DistilBERT / RoBERTa) to capture sarcasm and complex context, and implementing Aspect-Based Sentiment Analysis (ABSA).